Precedence and Associativity Operations - Zero to Hero Questions List

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1 Precedence and Associativity in C

1.1 1. Operator Precedence

Operator precedence decides **which operator is evaluated first** when multiple operators appear in the same expression.

1.1.1 Example

```
int x = 10 + 5 * 2;
```

Here:

```
5 * 2 = 10
10 + 10 = 20
```

Because `*` has **higher precedence** than `+`.

So:

```
x = 20
```

1.2 2. Associativity

Associativity decides the **direction of evaluation** when two operators have the **same precedence**.

It can be:

- Left to Right
- Right to Left

1.2.1 Left to Right Associativity

Example:

```
int x = 20 / 5 * 2;
```

Both / and * have same precedence.

Evaluation:

```
20 / 5 = 4
4 * 2 = 8
```

So:

```
x = 8
```

Because associativity is **left $\rightarrow$ right**.

1.2.2 Right to Left Associativity

Example:

```
int a, b, c;
a = b = c = 10;
```

Assignment operator works from:

```
right → left
```

So:

```
c = 10
b = 10
a = 10
```

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2 Important Precedence Table (Simplified)

Operators	Meaning	Associativity
`() [] -> .`	Function/Array/Member	Left $\rightarrow$ Right
`++ --`	Increment/Decrement	Right $\rightarrow$ Left
`* / %`	Multiplication	Left $\rightarrow$ Right
`+ -`	Addition/Subtraction	Left $\rightarrow$ Right
`<< >>`	Bitwise Shift	Left $\rightarrow$ Right
`< <= > >=`	Relational	Left $\rightarrow$ Right
`== !=`	Equality	Left $\rightarrow$ Right
`&`	Bitwise AND	Left $\rightarrow$ Right
`^`	Bitwise XOR	Left $\rightarrow$ Right
`|`	Bitwise OR	Left $\rightarrow$ Right
`&&`	Logical AND	Left $\rightarrow$ Right

Operators	Meaning	Associativity
| Logical OR | Left → Right | | | |?:| Ternary | Right → Left | | | |= += -= *= /=‘	Assignment	Right → Left

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3 Important Rule

3.1 Use parentheses () to avoid confusion

Example:

```
x = (a + b) * c;
```

Parentheses always get highest priority.

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4 Questions to Solve (With Hints)

1. Find the output of:

```
printf("%d", 10 + 5 * 2);
```

(Hint: `*` before `+`)

2. Find the output of:

```
printf("%d", 20 / 5 * 2);
```

(Hint: Same precedence → left to right)

3. Find the output of:

```
printf("%d", 5 + 2 * 3 - 1);
```

(Hint: Solve multiplication first)

4. Find the output of:

```
printf("%d", 10 - 2 + 3);
```

(Hint: `+` and `-` same precedence)

5. Find the output of:

```
printf("%d", 4 * 2 / 8);
```

(Hint: Left to right)

6. Predict the value of:

```c
int a = 5, b = 10, c;
c = a + b * 2;
```

(Hint: Multiplication first)

7. Find the output:

```c
int x = 10;
printf("%d", x++ + 5);
```

(Hint: Post-increment)

8. Find the output:

```c
int x = 10;
printf("%d", ++x + 5);
```

(Hint: Pre-increment)

9. Predict:

```c
int a = 2, b = 3, c = 4;
printf("%d", a + b * c);
```

(Hint: * first)

10. Find the result:

```c
int x = 5;
x *= 2 + 3;
```

(Hint: +=, *= precedence)

11. Find the output:

```c
printf("%d", 5 > 3 && 2 < 1);
```

(Hint: Relational before logical)

12. Predict:

```c
printf("%d", 5 + 3 > 6);
```

(Hint: + before >)

13. Find the output:

```c
printf("%d", 10 || 0 && 5);
```

(Hint: && before ||)

14. Predict:

```c
printf("%d", (5 + 3) * 2);
```

(Hint: Parentheses first)

15. Find the output:

```c
printf("%d", 8 >> 1 + 1);
```

(Hint: + before shift)

16. Predict:

```c
int a = 5;
int b = a++ * 2;
```

(Hint: Post-increment usage)

17. Find the output:

```c
int a = 5;
int b = ++a * 2;
```

(Hint: Pre-increment)

18. Predict:

```c
printf("%d", 5 & 3 == 1);
```

(Hint: == vs & precedence)

19. Find the output:

```c
printf("%d", 1 || 0 && 0);
```

(Hint: Logical precedence)

20. Predict:

```c
int a, b, c;
a = b = c = 5;
printf("%d", a + b + c);
```

(Hint: Assignment associativity)

21. Find the output:

```c
printf("%d", 10 - 5 - 2);
```

(Hint: Left to right)

22. Predict:

```c
printf("%d", 2 + 3 * 4 > 10);
```

(Hint: Arithmetic before relational)

23. Find the output:

```c
printf("%d", !5 + 2);
```

(Hint: Logical NOT precedence)

24. Predict:

```c
printf("%d", ~5 + 2);
```

(Hint: Bitwise NOT)

25. Find the output:

```c
printf("%d", 3 * 2 % 4);
```

(Hint: Same precedence → left to right)

26. Predict:

```c
int x = 5;
printf("%d", x += 3 * 2);
```

(Hint: Multiplication first)

27. Find the output:

```c
printf("%d", 5 == 5 || 0);
```

(Hint: Equality before logical OR)

28. Predict:

```c
printf("%d", 5 + 2 > 3 && 4);
```

(Hint: Relational before logical AND)

29. Find the output:

```c
printf("%d", 2 << 1 + 1);
```

(Hint: Addition before shift)

30. Predict:

```c
printf("%d", 4 + 5 * 2 == 14);
```

(Hint: Arithmetic before equality)